STATS 100A HONORS / PROFESSOR WU / OCTOBER 26, 2026

From noise to Chinese characters

First scope brief and mathematical illustration | ZiSheng project

Research question

Can a small class-conditional diffusion model generate recognizable images of selected Chinese characters? How do Gaussian corruption and character stroke count relate to recognizability?

Proposed study

Data. Begin with a controlled rendered-font pilot. Consider isolated handwriting for the main study only after verifying source access, labels, writer-level splits, character coverage, and use conditions. Keep the main analysis within one data domain.

Model. Condition a small denoiser on a character ID and timestep. Start with a three-character pipeline pilot; scale to at least nine characters if the data and training gates pass. Twenty-four is a target, not a commitment.

Evaluation. Assess requested-character identity separately from image fidelity. Treat stroke-count patterns as associations within the selected set, not evidence that stroke count causes performance differences.

Forward corruption: the first mathematical object

x_t = sqrt(alpha_bar_t) x_0 + sqrt(1 - alpha_bar_t) eps, where eps ~ N(0, I). Thus, x_t given x_0 is Normal with mean sqrt(alpha_bar_t) x_0 and covariance (1 - alpha_bar_t) I. The first expression constructs the noisy image; the second gives its conditional mean and covariance. This is the fixed forward process, not the learned reverse process.

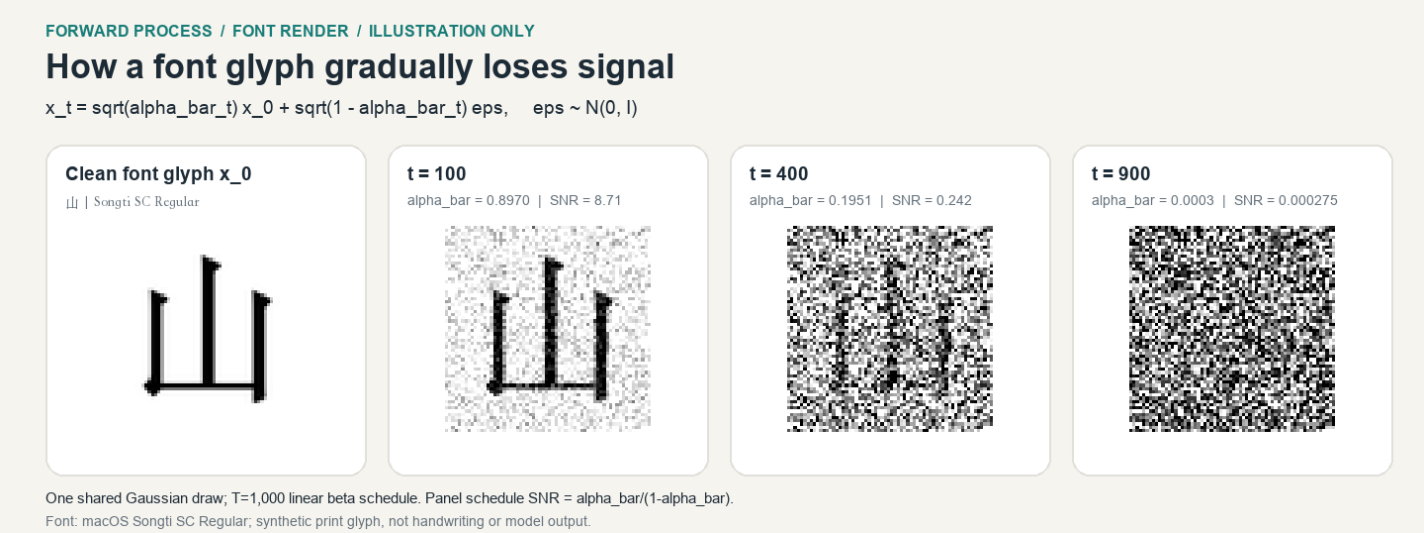


Figure 1. T=1,000 with a linear beta schedule from 0.0001 to 0.02; one shared seeded noise draw. Panel schedule SNR is alpha_bar/(1-alpha_bar). The glyph is rendered in macOS Songti SC Regular. Values are clipped to [-1, 1]. This synthetic font illustration is not handwriting, training data, or model output.

Status and next step

The repository has planning notes and interface placeholders only. No data have been acquired, and no model has been implemented, trained, or evaluated. Next: choose an openly licensed font for the actual dataset, verify character coverage and terms, then make a labeled contact sheet.

Questions for our meeting

1. Is rendered-font generation an acceptable controlled main domain if handwriting access or use terms delay the project?
2. Which outcome should lead: generated-character recognition or corruption/restoration of held-out images?
3. What report format, final deadline, and mathematical depth should I plan for?

Reference: Ho, Jain, and Abbeel (2020), *Denoising Diffusion Probabilistic Models*, Section 2, Equation 4. Open paper.